

Next-Gen Train Delay Forecasting: A Logical Approach with Neural Schema Optimization

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Abstract— The train delay not only disrupts the regular operation of the present schedules but also propagates to subsequent trains, leading to extensive delays across the network. Studying and predicting how train delays spread ahead of time can help railway transportation authority's control the number of delays and lessen the money lost due to them. The limited quantity of data samples available for train chains complicates the construction of train chain data. Recent studies that look at problems with train delays mostly use standard machine learning methods with small sample sizes. These systems frequently fail to encapsulate the intricate dynamics of train operations, which can fluctuate considerably according to variables such as meteorological conditions, maintenance timetables, and passenger demand. Thus, there is an increasing demand for more advanced methodologies capable of analyzing extensive datasets and integrating real-time data to enhance predictions and reduce disruptions. This study employs two probabilistic forecasting methods, specifically the optimal neural schema classification model (ONSC-M), to predict train delays in Indian Railways. The algorithms accurately anticipate the distribution of train delay arrivals and departures, achieving a mean absolute error of under 13 minutes. To demonstrate the applicability of the predicted delay distributions, we include these probabilistic forecasts in the probabilistic train-to-station assignment problem. The aim of this issue is to enhance the resilience of train-to-station assignments. Our suggested train-to-station assignment approach, taking into account probabilistic delay forecasts, decreases the number of conflicting trains by as much as 84% compared to a traditional train-to-station assignment model. The results demonstrate the value of employing probabilistic forecasting for the optimization of robust railway operations.

Index Terms—Train delay schedule, optimized neural classification model, probability distribution, Minimum Mean square error, probabilistic Neural network design.

I. INTRODUCTION

Alterations in process durations (including running and dwell times), capacity, and synchronization protocols may result in train delays. Their reliance may also hinge on the

accessibility of infrastructure, rolling stock, and personnel [1, 2]. Delays can be categorized as primary and secondary. An internal disruption within the process causes a primary delay, which extends the anticipated process length [3]. There are many things inside and outside the railway system that affect how long trains travel between stations and how long they stop. Because of this, it is important to look at how these things affect the network as a whole. Secondary delays occur as a result of first delays, frequently causing additional disruptions in the timetable and impacting future train services. There are three layers of railway traffic planning such as tactical, strategic and operational, these layers are established based on the planning horizon of railway operations [4]. At every planning level, a precise train delay prediction model serves a crucial function. The main objective of the train delay prediction to each tier of railway traffic planning and intricate on the applications of train delay prediction models at each tier. Strategic planning emphasizes the systematic design of the transport network and the long-term management of resources or vehicles to address future traffic demand, with resources encompassing adequate infrastructure capacity, rolling stock, and train personnel to support anticipated traffic flows [5].

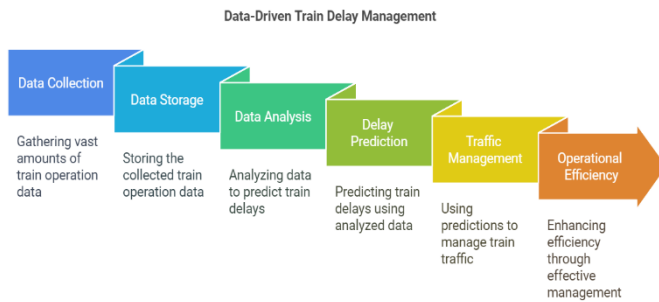


Fig.1 Data-Driven train delay management protocol

In [6], data-driven train delay prediction models are categorized into global and local models according to their spatial forecasting range. In [7], the models are categorized into long-term and short-term delay prediction models according to their temporal forecasting capabilities. The global model and long-term train delay prediction model, as well as the local model and short-term train delay prediction model, are classed by different criteria but are equivalent and perform the same functions. Therefore, this chapter only utilizes the terms global and local models to eliminate ambiguity. The principal advantage of global regression models is their result generalizability, enabling the applicability of findings from the global model to train lines absent from the training datasets [7,8]. Figure 1 depicts the efficient data-driven techniques for forecasting train delays. These strategies include diverse statistical techniques and machine learning algorithms that improve the precision of delay estimates across varied train networks. In this researchers we proposed ONSC- Model by utilizing broad datasets to adapt to evolving situations and increase operational efficiency in the railway transportation sector.

II. RELATED STUDY

A wide range of researchers has undertaken broad ranging of investigation on the subject of train delays. In [9] determined the complete postponement period of a train on a confident route by analyzing the temporal acceptable time and the successively time of the trains arrive in delayed manner, utilizing a closed analytic formula grounded in delay propagation theory and validating the findings through simulations using software. In [13], the researchers developed a modified framework, working a maximum statistical model, erected a unique matrix with the delays of trains together and to the getting delay train are eigenvalues component, used the optimal control protocol to establish the model factors, and tested the train delay based on the awaiting connection. Postpone the train to assess the delay over the entire transportation network. The author of [10] created the *GaM* {1,2} methodology to predict train delays. They utilized the “Markov matrix” to fix errors and the radial basis function (RaB-F) as the neural network layers and to intercalate and enlarge the data while diminishing the relative error. The researcher [11] employed heuristic methods to derive the operational plan following train disturbances and

delays, optimizing speed and positioning to minimize overall train delays within the study parameters. In [12], the author introduced an innovative multitrain trajectory optimization method for solitary track lines. They focus on delay instances intended to identify ideal speed profiles that minimize delays and conserve energy consumption. Nevertheless, some critics contend that an excessive dependence on heuristic methods may result in inefficient solutions, as they could neglect essential aspects that influence overall system performance. Moreover, although trajectory optimization for isolated track lines is advantageous, it may prove less effective in intricate train networks where numerous elements and interactions are involved.

Developing data-driven train delay prediction models is a complex endeavor that encompasses problem definition, source and sink representations, preprocessing of data, learning modeling techniques, model evaluation, and particularly, the inter-protocol procedure of predictive analytics, necessitating expertise from both computer science and railway sectors [13]. Additionally, other critical technical issues pertaining to each facet require resolution, including the methodology for processing data from multiple sources prior to its integration into the predictive model. Given that effective solutions for various data-driven train delay prediction modeling issues are dispersed across existing studies, a thorough synthesis of validated techniques from current research not only provides a valuable reference for railway researchers encountering analogous modeling challenges but also offers contemporary methods for the relevant modeling components [14]. The absence of insights from current studies impedes the rapid adoption of data-driven methodologies, hence constraining advancements in train operation management methods [15]. To summarize, the data driven based predictive model having the complexity of the multistage train delay prediction, to overcome these issues in this research we proposed the Optimized neural model for efficient prediction of the multistage train delay occurrences.

III. PROPOSED METHODOLOGY

This proposal suggests a multi-stage strategy for predicting train delays, utilizing data smoothing and multimodal synthesis. The initial phase of the multi-stage framework involves to segregate the train execution data details based on isolating the normal components and residual components from the unwanted data. The second step employs Optimized neural network model to train the flat data and uses the Schema categorization for trainers (ONSC-M). The chronological average residual value serves as the predictive residual component, and ultimately, the predictive values of the smooth and residual components are amalgamated. Consequently, the framework incorporates a data pretreatment phase into the standard prediction model and utilizes the synthesized outcome of the smoothed residual value alongside the proposed optimized model.

We are using Indian Railways' historical database in our proposed study, recording the current values at the

forecast time. In Indian Railways, data on passenger train delays is recorded and documented daily. Over the weekend, Indian Railways aggregates all individual daily data to create the weekly dataset. We consolidate all weekly data at the end of the month to generate the monthly delay value. This monthly delay metric offers a thorough assessment of train performance over time, facilitating enhanced analysis and predictions. By aggregating these facts, we can discern trends and patterns that may guide our proposed prediction model enhancements in service efficiency. These data allow us to execute focused tactics designed to minimize delays, elevate customer satisfaction, and eventually enhance the overall reliability of the railway system. Moreover, systematic evaluations of this data might facilitate the modification of operating protocols and the more efficient allocation of resources.

Table. I Passenger Train Delay Timings

YEAR	No of trains dispatched	Total Delay [in Mints]	Total Delay Average (Mins/Train)
2019	28927	666468	23.04
2021	31121	738293	23.86
2022	34171	1334081	39.28

In Table .I, the average delay of the each train was calculated to divide the total delay by the number of dispatched trains. For this above table .I we find the maximum delay occurred in the year of 2022 nearly 40Mins of average delay mints of each train.

The proposed model is exemplified in Figure 1. In the designed framework of train delay analysis, consider two types of data sequences: chronological data (x_d) and current observation data (x_L) pertaining to train operations. Historically recorded data is utilized for our training procedure, while present observational data indicates the present status of train operations. During the initial data cleaning phase, the historically recorded data x_h is segmented into a value added component x_{df} and it's residual x_{dr} in the offline procedure. The residual sequence x_{dr} is concurrently defined by the reordered data's mean of the residual $\hat{x}_{dr}$. In the connected technique, the x_{Lf} component within the observational data is refined using data smoothing extraction. The ultimate prediction outcome is the aggregate of the forecasted smoothing $\hat{x}_{Lf}$ and the assessed residual $\hat{x}_{dr}$ derived from previous data. Machine learning techniques are incapable of accurately forecasting actual asymmetrical data patterns. Consequently, the framework exclusively employs the Proposed ONSC-Model to forecast the continuous sequence, whereas the residual component utilizes average analysis.

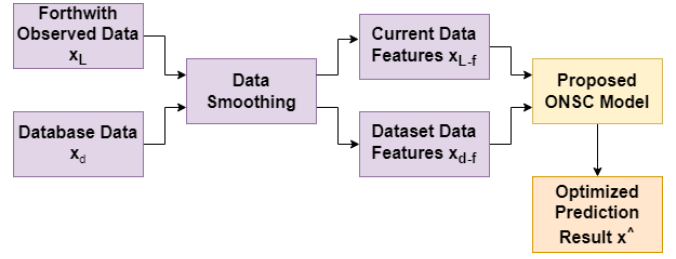


Figure.2 Proposed Model Block Diagram

The suggested ONSC-Model integrates a neural network with a model combined with the Gaussian distribution. For the data essential feature values x_{Lf_i} of train i , an ONSC-Model generates the Gaussian based parameter of the weight value α , center point of data mean m_μ , and the distribution of data in the form of standard deviation SD_σ . The probability density function $p(y_{tj}|x_{si})$ of the target variable y_{tj} , representing train delay, is established using these parameters. The ONSC-Model is especially adept at estimating multimodal probability distributions. It can consequently estimate the entire process local and global maxima in the middle of two distinct likely delay values. The following weighted sum of Gaussian distributions represents the train delay probability distribution:

$$p(y_{tj}|x_{si}) = \sum_{j=1}^m \alpha_j(x_{si}) \phi_j(y_{tj}|x_{si}) \quad (1)$$

$$\phi_j(y_{tj}|x_{si}) = \frac{1}{\sqrt{2\pi\sigma_j(x_{si})^2}} \exp\left(-\frac{(y_{tj} - \mu_j(x_{si}))^2}{2\sigma_j(x_{si})^2}\right) \quad (2)$$

In this context, $p(y_{tj}|x_{si})$ denotes the probability distribution of the delay value y_{tj} conditioned on the feature values x_{si} from train sample i . Additionally, the weight of $\alpha_j(x_{si})$, local maxima parameter $\mu_j(x_{si})$, and finally the distribution details parameter $SD_\sigma = \sigma_j(x_{si})$ of the j^{th} Gaussian factor, where $1 \leq j \leq m$, with m indicating the total number of Gaussian components included in the combination procedures.

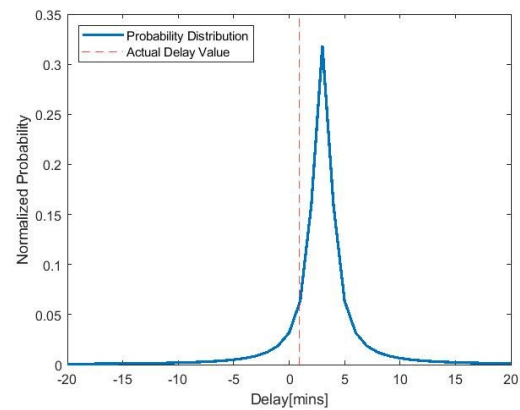


Figure.3 Model Delay Distribution Function

Figure 3 illustrates a scenario of the probability distributions derived using the techniques suggested. The real latency value for the train instance is specified for distributions.

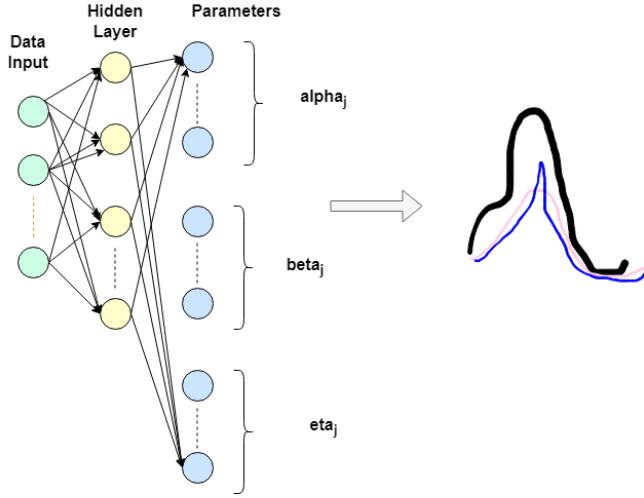


Figure .4 Schematic representation of a proposed ONSC-Model

The major focus on the training procedure undertaken depends on the back propagation, wherein the network parameters, explicitly the biases and the weights of each node, are adjusted based on an error computing function E_r , defined as the negative logarithm of the likelihood that the model produced by the current network sink, then generates the trained data model. This probability is the product of the likelihoods of each data value, conditioned on the obtainable network parameters. In a formal manner,

$$E_r = \sum_{i=1}^{M_f} \left(-\ln \sum_{j=1}^n \alpha_j(x_{si}) \phi_j(y_{ti}|x_{si}) \right) = \sum_{i=1}^{M_f} -\ln p(y_{ti}|x_{si}) \quad (3)$$

Algorithm 1 – Proposed ONSC- Model

1. Input: parameters of X, e_f, m, λ_s
2. Output: $RMSE_{oM}$ and optimized cdf_{oM}
3. Create multidimensional data T_{X_m} as

$$T_{X_M} = \sum_{j=1}^p \mu_j \sigma_j v_j^T$$
4. Grouping and extending the T_{X_M} as

$$T_{X_M} = f_{T,SVD}(T_1, T_2, \dots, T_p)$$
5. To create additive components using diagonal Averaging
6. Calculate Energy function of the model

$$e_f(w_f, b_f, y, x_j) = (y_f - \hat{t}_n)^2 + \frac{1}{M-1} \sum_{j=1}^{M-1} (y_o - \hat{x}_j)^2$$
7. Enable the Attention procedure

$$T_{X_M} = \text{SigModi} \left(\frac{1}{E_f} \right) \otimes T_{X_M}$$

8. Calculate $RMSE_{oM}$ and Optimized cdf_{oM}
Using equation (4) and (5)
7. End of the Algorithm

The Algorithm 1 shows the proposed model execution procedure. In this step 3 to step 5 is used to data smoothing and feature selection of both historical and current observed train arrival to the target station with the reference information of previous station departure time. The one dimensional data X is transformed as multidimensional data using the SVD trajectory matrix and produce the T_{X_M} . The parameters μ_j, σ_j and v_j^T are Eigen vectors of left and right trajectory matrix produced by SVD decomposition. The variable p is the rank of the matrix.

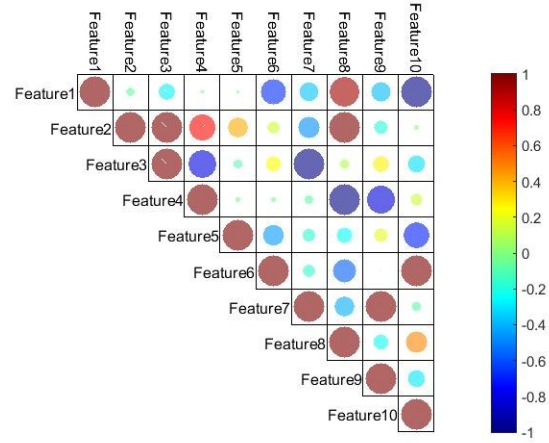


Figure.5 Multidimensional Feature Selection for Proposed Model

$$RMSE_{oM} = \sqrt{\left(\frac{1}{M_f} \right) \sum_{j=1}^{M_f} (y_{tj} - \mu_M(x_{si}))^2} \quad (4)$$

$$cdf_{oM} = \frac{1}{M_f} \sum_{j=1}^{M_f} |(\hat{y}_{tj} - \mu_M(x_{sj}))| \quad (5)$$

To assess the significance of each neuron, the simplest method is to differentiate the target neuron from the other neurons using the Energy function step. 6 is delineated in Algorithm 1. In this context, $\hat{t}_n$ and $\hat{x}_j$ denote the target neurons and the other neurons of the input three-dimensional eigenvector x_j , which signifies the index in the spatial dimension. M represents the total number of neurons within

a channel, while w_f and b_f correspond to the weight and bias of neurons during transformation.

Given that the traditional RMSE and CDF are inadequate for evaluating an estimated train delay distribution, the suggested ONSC-Model generates the variants $RMSE_{OM}$ and cdf_{OM} using equations (4) and (5), which are derived by contrasting the mean of the estimated distribution with the actual delay value. During the process of estimating multimodal probability distributions, the ONSC-Model becomes more accurate. Because of this, it is able to make predictions about a distribution that has peaks at, for instance, two different probability delay values. Due to the presence of this feature, the model is able to provide more accurate projections, which are reflective of the complex realities of train schedules.

IV. RESULTS AND DISCUSSION

This section presents the suggested ONSC-Model, which is based on the modified neural network. We perform independent prediction experiments for the departure train delay levels at level 3 railway stations, influenced by the propagation of prior train delays. The experiment demonstrates the prediction of departure delay levels at level 3 railway stations influenced by the departure delays of level 1 and level 2 railway stations. So, the labels for the training dataset are the train delay classifications at level 3 railway stations. The test dataset is used to make predictions about flight delay levels. The following part presents the data, experimental environment, and parameter configuration, followed by a comparison of the model's performance before and after enhancement using several metrics.

Figure 6 illustrates the dataset employed for the experiments in this research, comprising Indian railroads data from the years 2019, 2021, and 2022, sourced from a third-party provider. Train chain data are generated from the train movement data recorded three times within a specified time interval. The first dataset utilized in the trials comprises 1,048,576 entries. The original data categorizes five categories of delay based on delay duration, with the proportions of no delay, minor delay, moderate delay, high delay, and severe delay being 64:21:7:5:4.

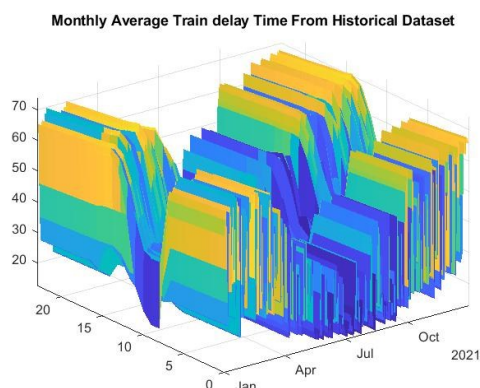


Figure .6 Single year (2021) dataset heat map

Following data cleansing and the building of the train chain dataset, the final dataset utilized for analyzing train delay propagation comprises 35,376 data entries. Subsequent experiments are performed and validated on this train chain dataset. The training set and validation set are partitioned at a 4:2 ratio.

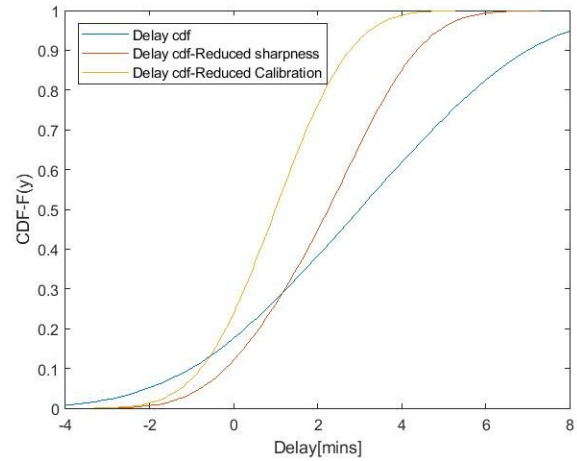


Figure. 7 Depicts the correlation

In Figure.7, The stepping function at the actual delay value (blue line) aligns with an anticipated value of zero minutes without delay. The yellow and orange lines represent the same cumulative distribution function with diminished sharpness and calibration, correspondingly. In figure.8 shows the proposed protocol prediction accuracy in terms of the cumulative distribution of the multistage data. The green lines shows the actual expected prediction of train arrival from the class 3 levelling railway station with respect to the class 2 (Previous) railway station departure.

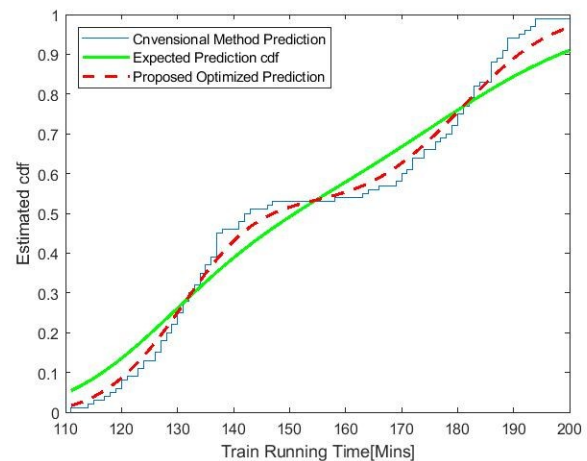


Figure.8 Proposed Model Prediction efficacy in terms of estimated CDF.

The total distance will be calculated towards the train running time. The final proposed model outcome (dotted red line) shows this model outperforms comparing the conventional prediction methods.



Figure.9 Proposed ONSC-Model validation performance with respect to epochs

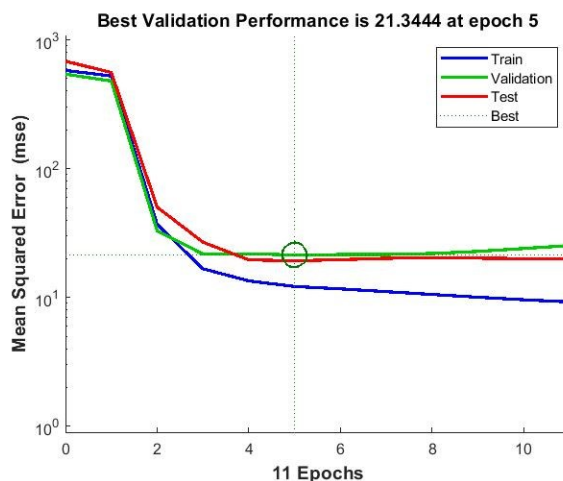


Figure.10. Proposed Model validation with respect to MSE_{oM}

Model validation is a more essential procedure in the development of machine learning models and also important for guaranteeing that the model functions properly and follows to ascendancy requirements. The Figure.10

rationalize our proposed ONSC- Model outperformed the prediction train delays using the historical trained data with the current observed data, also the validation value shows that our model greatly diminishes the intricacy of the model.

V. CONCLUSION

The ONSC-Model improves in estimating multimodal probability distributions. It can therefore predict a distribution with peaks at, for example, two separate likely delay values. This functionality enables the model to deliver more precise forecasts, mirroring the intricate realities of train schedules. By considering the unpredictability and uncertainty in train delays, it provides useful insights for operational planning and passenger communication. The ONSC-Model may ascertain various probable outcomes for train delays, rather than merely a singular average forecast. Consequently, it improves decision-making by offering a more sophisticated perspective on potential possibilities. The results will demonstrate the advancements attained through the enhancements, emphasizing the efficacy of the offered procedures. Furthermore, we will examine the ramifications of these findings for operational efficiency and prospective research avenues in train transportation analytics.

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